

Give the Model Data, Give the User a UI

MCP Apps

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Practical MCP Apps

Lessons from the trenches

- Claude
- ChatGPT
- Indeed Career Scout

I'm looking for a Barista job in Austin, TX.

I can help you search for that. Let me check Indeed for current openings.

Loaded tools, used Indeed integration

Here's what's currently open in Austin and nearby:

In Austin proper:

- **UT Campus Barista - Lucky Lab Coffee Co.** — Part-time, \$16-\$21/hr, posted June 19
- **Barista - Caffe Medici** — Full-time, \$15-\$25/hr
- **Barista (FT/PT) - Morrison Healthcare** — Full-time, \$21-\$24/hr, posted June 16
- **Barista (FT/PT) - Eurest** — Full-time, \$16-\$17/hr, posted June 18

Nearby (Cedar Park, Leander, Round Rock):

- **Barista - Bagel Bear** — Part-time, \$10-\$20/hr, Cedar Park
- **Barista - Summer Moon Coffee Bar** — Part-time, \$16-\$20/hr, Leander
- **Barista, Server - Lumi's Crepes** — Part-time, \$12-\$18/hr, Cedar Park
- **Starbucks Barista - Target (T1066)** — Part-time, from \$16.50/hr, Round Rock
- **Barista - Alamo Coffee Company Texas** — Part-time, \$16-\$20/hr, Round Rock

I'm looking for a Barista job in Austin.  Indeed

 Indeed



Carpenter Hotel
★ 3 • 2 days ago

Full Time Barista- Carpenter Hotel

Austin, TX

Pay **Not specified**

Job type **Full-time**

Setting **Not Specified**

Summary
The Opportunity:
A Barista will exceed the expectations of...

Apply on Indeed

View details



Marriott International
★ 4 • 3 days ago

Barista

Austin, TX

Pay **Not specified**

Job type **Full-time**

Setting **Not Specified**

Additional Information Job
Number26078131 Job CategoryFood and
Beverage & Culinary Location9721...

Apply on Indeed

View details



Thor
★ 2 •

The Coffe Time

Austin, TX

Pay

Job >

Setting

Summary
The Role
At Hyatt, we

Apply on In

Good cheap restaurants near tamachii station in Tokyo.

Searching for places



The map shows the area around Tamachi Station in Tokyo. Several restaurants are marked with their ratings: 4.5, 4.4, 4.3, 4.2, 4.0, and 3.8. The map includes labels for 'Keidaiseimon 慶大正門前', 'SHIBA 芝', 'SHIBAURA 芝浦', '3 CHOME 3丁目', and 'Nagisa Dori'. It also shows major roads like Route 301, Route 409, and Route 15. The Google logo is visible in the bottom left corner of the map area.

Cheap Eats near Tamachi Station

-  **Shinpachi Shokudo...**
4.0 ★ (1,374) · Japanese ...
Authentic Japanese set meals & grilled fish from...
-  **Masaru**
4.4 ★ (1,260) · Yakiniku R...
Beloved local yakiniku (BBq beef). Often has a queue ...
-  **Kushikatsu Tanaka...**
4.5 ★ (965) · Japanese Iz...
Deep-fried skewers chain.
5-min walk from station. ...

Map data ©2026 Terms Report a map error

Here are some solid cheap eats near Tamachi Station, all budget-friendly (¥ price level):

 **Shinpachi Shokudo Tamachi** ★4.0 (1,374) ↓ ws)

Create a sequence diagram for an MCP call. Use Excalidraw.

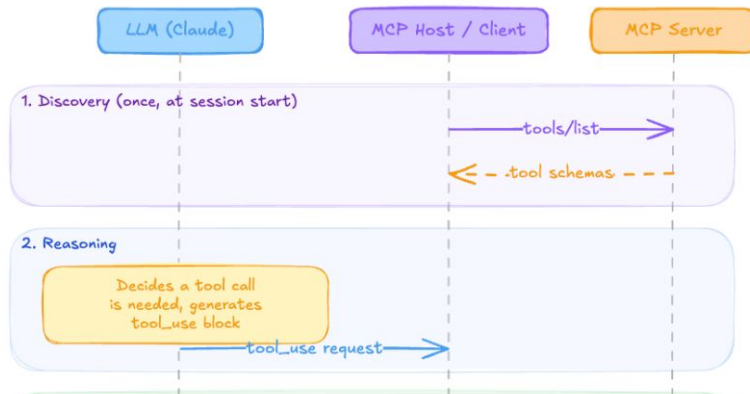
Let me load the Excalidraw format reference first.

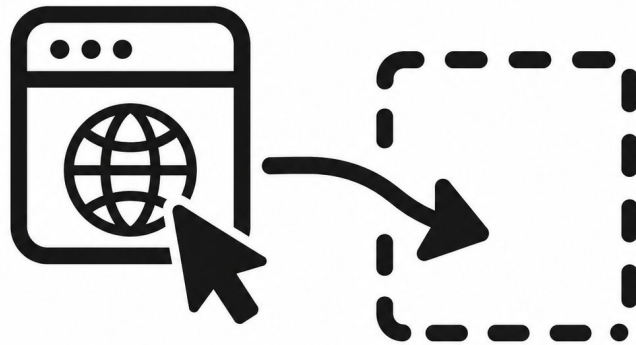
Loaded tools, used Excalidraw integration

This is helpful — there's actually a near-identical example already in the reference (MCP Apps sequence flow). I'll adapt it to a standard MCP tool-call sequence: **Client (Claude) → MCP Host/Client runtime → MCP Server**, showing the initialize/list-tools/call-tool flow that's core to any MCP call.

✂ Excalidraw

MCP Tool Call — Sequence Diagram





I'm looking for a Barista job in Austin.  Indeed

 Indeed



Okay 

???

Huh?

IDK

Tell me about the first result

Which of these are in-person?

Please rank the companies

Rule #1

Anything shown to the user should be provided as data to the model

```
27 registerAppTool(server,
28     "get-time",
29     {
30         title: "Get Time",
31         description: "Returns the current server time as an ISO 8601 string.",
32         inputSchema: {},
33         outputSchema: z.object({
34             time: z.string(),
35         }),
36         _meta: { ui: { resourceUri } }, // Links this tool to its UI resource
37     },
38     async (): Promise<CallToolResult> => {
39         const time = new Date().toISOString();
40         return {
41             content: [{ type: "text", text: time }],
42             structuredContent: { time },
43         };
44     },
45 );
```

Path to UI



**Data as
structuredContent**



Austin, TX

Pay

Not specified

Job type

Full-time

Setting

Not Specified

Summary

The Opportunity:
A Barista will exceed the expectations of...

Apply on Indeed

View details

Austin, TX

Pay

Not specified

Job type

Full-time

Setting

Not Specified

Additional Information Job
Number26078131 Job CategoryFood and
Beverage & Culinary Location9721...

Apply on Indeed

View details

Austin, TX

Pay

>

Job

Setting

Summary

The Role
At Hyatt, we

Apply on In

Thought for 4s >

I found **90+ barista jobs around Austin** on Indeed.

A few patterns worth noting:

- There are both **full-time and part-time** roles.
- Pay is listed on some postings, with examples around **\$10/hr + tips**, **\$12–\$13/hr + tips**, and **\$16.25–\$18.44/hr**.
- Several roles are at hotels/cafes, Starbucks locations, Target Starbucks, grocery-store Starbucks, and neighborhood coffee shops.
- Some postings require weekend/holiday flexibility; a few mention food handler or TABC certification.

The results are shown above with apply links.

Tool Description

... Results are automatically displayed to the user as UI components.

I'm looking for a Barista job in Austin.  Indeed

 Indeed



Carpenter Hotel

★ 3 • 2 days ago

Full Time Barista- Carpenter Hotel

Austin, TX

Pay **Not specified**

Job type **Full-time**

Setting **Not Specified**

Summary

The Opportunity:

A Barista will exceed the expectations of

Apply on Indeed

View details



Marriott International

★ 4 • 3 days ago

Barista

Austin, TX

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Additional Information Job

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Apply on Indeed

View details



Thor

★ 2 •

The Coffee Time

Austin, TX

Pay

Job

Setting

Summary

The Role

At Hyatt, we

Apply on Indeed

???

Huh?

IDK

Summarize this job description

Create a cover letter for this job

Save the applied jobs to my log

Rule #2

**Information about any user interaction
should be provided to the model**

Copy

```
const markdown = `---  
item-count: ${itemList.length}  
total-cost: ${totalCost}  
currency: ${currency}  
---
```

User is viewing their shopping cart with `${itemList.length}` items selected:

```
${itemList.map((item) => ` - ${item}`)}.join("\n")}`;
```

```
await app.updateModelContext({  
  content: [{ type: "text", text: markdown }],  
});
```


Afternoon, Dustin

I'm looking for an AI Engineer job. I'm willing to relocate. I'm looking for the highest-paying options. I don't want to work in the defense, finserve, or healthcare industries.

+

Sonnet 4.6 High ▾



Job Search

Better — this gives some real signal (note: Airwallex is fintech/payments, so I'm flagging that as borderline for your finserve exclusion). Let me also check NYC and Seattle, since comp at the top end often clusters there too, and run one more SF query targeting "Principal" level given your background.

Used Indeed integration

Filtering out the defense (Deloitte's secret-clearance role), finserve (Gesa Credit Union, Airwallex), and excluding T-Mobile/Microsoft as borderline-fine but not top comp — here's where things stand. Given your seniority (Principal/Technical Fellow level), most of these "Senior" listings are actually a level or two below where you'd land comp-wise. Let me get one more pull from NYC, since that's a major hub for AI eng comp outside SF, before I summarize.

Job Search

One flag worth noting: **Anterior** does clinical/healthcare-claims AI, so I'm excluding it per your filter even though the comp (\$230k-\$300k) is strong.

Here's the consolidated, filtered list of the strongest non-defense/non-finserve/non-healthcare options found so far, sorted by top of comp range:

Role	Company	Location	Pay Range
<u>Software Engineer, AI Dev Tools</u>	Whatnot	San Francisco, CA	\$195,000 – \$290,000
<u>Software Engineer – Deepgram for Restaurants</u>	Deepgram	San Francisco, CA	\$154,000 – \$307,000

Rule #3

Separate data processing from UI rendering

Home

Quickstart

Core Concepts

MCP Apps in ChatGPT

MCP Server

UX principles

UI guidelines

Plan

Research use cases

Define tools

Design components

Build

Set up your server

Build your ChatGPT UI

Authenticate users

If you need both read and write behavior, create separate tools so ChatGPT can respect confirmation flows for write actions.

Separate data processing from UI rendering

If one workflow needs both reusable data and a widget, plan that as two tools instead of one overloaded tool:

- **Data tools** return complete `structuredContent` for model reasoning and follow-up calls, without a UI template.
- **Render tools** accept the prepared data, attach the component template, and stay focused on presentation.

The model should call the data tool first, use the returned `structuredContent`, then call the render tool with the prepared data so the widget renders once with final, model-checked context. State that dependency in the render tool description.

For local UI interactions that need fresh data, let the widget call the data tool directly rather than remounting itself. See [Build your ChatGPT UI](#) for the fuller implementation pattern.

Capture metadata for discovery

Discovery is driven almost entirely by metadata. For each tool, draft:

- **Name** – action oriented and unique inside your connector (`kanban.move_task`).
- **Description** – one or two sentences that start with “Use this when...” so the model knows exactly when to use the tool.

Ask AI

```
search_jobs(title, location)
```

|
v

```
search_jobs(title, location)  
render_jobs_widget(job_ids)
```

```
// Description: ... Always call `search_jobs`  
first ...
```

Key Takeaways for MCP Apps

Focus on the data before the UI

Small, composable tools

Thanks!

